

Report # IR-09

AUDIT REPORT

GENERAL

TUV COORDINATORS: RAMY MAREI MOHAMED EL-SAYED	TÜV INSPECTOR: DIMITRIS NALBANTIS
TYPE OF VISIT: ☒ IN-PROCESS INSP ☐ PARTIAL FINAL INSP. ☐ FINAL INSP. ☐ ORDER STATUS	

CLIENT/PROJECT INFORMATION

CLIENT: FOUR SEASONS HOTEL NILE PLAZA	CONTRACT NUMBER: PO NUMBER :	20BISIEG0109 CAI0091143
CONTACTS: Mohamed Abdel Latef, Director of Engineering Ahmed El Bashary, Maintenance Manager Mostafa Mahmoud, Chief Engineer	PROJECT No. :	3/2021

SUMMARY

DATE OF LAST VISIT: THIS IS THE 9 TH REPORT DATE(S) OF THIS VISIT: THIS IS A "REMOTE" DATA AUDIT ON APRIL 2022 DATA CONDUCTED ON 20 MAY 2022
NEXT SCHEDULED VISITS: THE NEXT SCHEDULED ONSITE VISIT IS PLANNED FOR THE END OF MAY 2022
NON-CONFORMITY REPORT ISSUED DURING THIS VISIT: ☒ NO ☐ YES NCR NO.: -
OUTSTANDING NON-CONFORMITY REPORT: ☒ NO ☐ YES NCR NO.:
INSPECTION RELEASE ISSUED DURING THIS VISIT: ☐ NO ☐ YES RELEASE NO.: NOT APPLICABLE
ACTION POINTS: (NCR'S, clarifications, pending approvals, etc.) : SEE RECOMMENDATIONS & ACTION PLAN IN REPORT AND ATTACHED

AUDIT R E P O R T

FOUR SEASONS HOTEL, Nile Plaza, Cairo, Egypt
DEVELOPMENT AND IMPLEMENTATION OF A MONITORING PROGRAM TO EVALUATE THE
PERFORMANCE OF THE UTILITY SYSTEMS
REPORT ON THE 9TH "REMOTE" AUDIT OF DATA FOR APRIL 2022
DATE OF REPORT: 20/05/2022

1. TASK

The agreed upon task of the service is:

Development and implementation of a monitoring program to evaluate the performance of the utility systems at Four Seasons Hotel, Nile Plaza, Cairo.

2. IMPLEMENTATION OF THE TASK FOR 2021-2022 (monitoring program)

The implementation of the task of developing and implementing a monitoring program in 2021-2022 includes:

- "ON SITE" audit visits to the Hotel (4 times in a 12 month period)
- "REMOTE" audit reviews of data (data for 8 months in a 12 month period)

The duration of the "on site" audit visit will be 2 working days (10 hours on site and 6 hours for writing the report).

During the on-site visits and/or the remote data reviews, critical parameters will be checked and comments/recommendations will be issued. A written report will be submitted for each audit, within 5 working days of the visit and/or receiving the data.

Systems included in the scope of the audits:

- Cooling Towers & side-stream filters
- Hot water boilers and hot water closed systems, including heat exchangers
- Chillers and chilled/cold water closed systems, including heat exchangers
- Mains Water, potable and laundry
- Steam generator
- Swimming Pools
- General issues related to water treatment

AUDIT R E P O R T

3. PLAN OF THIS 9TH “REMOTE” AUDIT

The 9th “remote” audit on data focuses on the following subjects:

- System Technique (ST) four Service Visit Reports (SVR) for April 2022
- System Technique (ST) Chemical consumption table for Jan-April 2022
- System Technique (ST) Data for Chlorination, Turbidity and soluble Iron in the Cooling Towers and Alkalinity, soluble Iron in the Hot Water Boilers for December 2021 to April 2022
- Four Seasons (FS) Makeup water consumption data for April 2022

All above documents received by e-mail on 15th May 2022.

4. SUMMARY OF CONCLUSIONS / RECOMMENDATIONS

The sections below refer to the above issues

A. PROGRESS ON RECOMMENDATIONS of THE AUDIT VISITS

Attached you will find the updated comprehensive table depicting the progress achieved on the issues which were raised:

- In the onsite visit in August 2021
- In the onsite visit in November 2021
- In the onsite visit in February 2022

The progress is shown in the last columns “Auditor 17/4/22” and “Feedback 1/5/22”.

After the next onsite audit in the end of May 2022, the “closed” issues will be removed from the table, while the “pending” issues will remain.

The Table was part of the body of the Report, but due to size it is now attached as Attachment 1 – Action Plan – Comments & Feedback

Comments

- Progress has been made on many issues

AUDIT REPORT

Recommendations

- The next visit priorities are clearly indicated in the last columns of the table

B. COOLING TOWER MASS BALANCE – WATER CONSUMPTION

The following data were submitted for the water consumption in February and March 2022. Averages and standard deviations were calculated for the daily water consumption by the auditor.

FEBRUARY 2022 - Water consumption for cooling towers				Chilled Water		Oil Boiler		B2 Cooling tower	
	Main, m3/d	Kitchen, m3/d		m3/d		m3/d		New, m3/d	
Total cons.	844	170		0		81	0	251	
AVERAGE	30,1	6,1		0,02		2,9		9,0	
STD DEVIATION	17,8	3,8				1,6		8,8	

MARCH 2022 - Water consumption for cooling towers				Chilled Water		Oil Boiler		B2 Cooling tower	
	Main, m3/h	Kitchen, m3/h		m3/h		m3/h		New, m3/h	
Total cons.	1903	230		4		87	0	697	
AVERAGE	61,4	7,4		0,1		2,8		22,5	
STD DEVIATION	32,8	4,3				1,0		7,7	

Water consumption for cooling towers				Chilled Water		Oil Boiler (Steam)		B2 Cooling tower	
Total cons.	4275	328		3		84	0	665	
AVERAGE	142,5	11,3		0,1		2,8		22,2	
STD DEVIATION	46,5	10,0				1,7		9,7	

Comments

- Cooling Towers: The water consumption in all the cooling towers increased from February to April, as expected, as the ambient temperature increases and the heat load increases

AUDIT REPORT

- Cooling Towers : The standard deviation of the daily water consumption is high in all cooling towers indicating a high variability of this parameter
- Chilled water : The monthly losses in chilled water systems are very low
- Oil Boiler = Steam Boiler : the average daily water consumption is stable, indicating a stable load (assuming that the condensate return is stable).

The above data need to be combined with the cooling tower mass balance and cooling tower efficiency calculations as described in previous reports.

As per the attached Action Plan, these issues (including the issue of water consumption versus chemical consumption) will be discussed in detail and calculations will be made on site in the next visit.

C. CORROSION MONITORING THROUGH CORROSION COUPONS

There are no reported data on the corrosion rate after the first series of coupons, most probably because the coupons are not yet removed from the system.

D. COOLING TOWERS – TURBIDITY, CHLORINATION, SOLUBLE IRON

The following data were submitted for the period Dec 2021 to April 2022.
Averages were calculated by the auditor.

Main Cooling Tower					New Cooling Tower					Kitchen Cooling Tower				
Visit date	Iron (Soluble)	Turbidity In	Chlorine Supply	Chlorine return	Visit date	Iron (Soluble)	Turbidity In	Chlorine Supply	Chlorine return	Visit date	Iron (Soluble)	Turbidity In	Chlorine Supply	Chlorine return
05-Dec		7	0,4	0,27	05-Dec		5	0,4	0,28	05-Dec		7	0,44	0,31
12-Dec		6	0,41	0,29	12-Dec		6	0,42	0,32	12-Dec		7	0,45	0,31
21-Dec		7	0,45	0,3	21-Dec		7	0,41	0,34	21-Dec		6	0,42	0,3
27-Dec		8	0,46	0,3	27-Dec		6	0,44	0,27	27-Dec		6	0,4	0,3
03-Jan		7	0,41	0,28	03-Jan		7	0,47	0,29	03-Jan		7	0,41	0,32
09-Jan		9	0,44	0,32	09-Jan		8	0,45	0,32	09-Jan		8	0,45	0,34
17-Jan		7	0,47	0,34	17-Jan		7	0,5	0,35	17-Jan		7	0,46	0,32
23-Jan		8	0,4	0,27	23-Jan		out of order	out of order	out of order	23-Jan		10	0,43	0,29
31-Jan		12	0,4	0,2	31-Jan		out of order	out of order	out of order	31-Jan		18	0,41	0,3
07-Feb		15	0,39	0,2	07-Feb		out of order	out of order	out of order	07-Feb		17	0,37	0,28
16-Feb				0,27	16-Feb				0,25	16-Feb				0,32
20-Feb				0,25	20-Feb				0,33	20-Feb				0,32
28-Feb				0,2	28-Feb				0,22	28-Feb				0,21
06-Mar				0,2	06-Mar				0,15	06-Mar				0,2
14-Mar				0,21	14-Mar				0,4	14-Mar				0,4
21-Mar				0,22	21-Mar				0,36	21-Mar				0,35
28-Mar	0,08			0,17	28-Mar	0,10			0,35	28-Mar	0,12			0,37
03-Apr				0,2	03-Apr				0,4	03-Apr				0,4
10-Apr				0,2	10-Apr				0,33	10-Apr				0,35
17-Apr				0,21	17-Apr				0,36	17-Apr				0,4
26-Apr	0,1			0,23	26-Apr	0,11			0,33	26-Apr	0,14			0,44
AVERAGES	0,09	9,7	0,42	0,23	AVERAGES	0,105	7,3	0,47	0,32	AVERAGES	0,13	11,2	0,42	0,33

AUDIT R E P O R T

Comments

- Turbidity : it is under control, very low values in all cooling towers
- Chlorination :
 - o The chlorine level in the supply lines is sufficient (about 0,4 ppm in all 3 cooling towers), but there is recorded data for the "Supply" concentration of free chlorine
 - o Similar comment as in previous reports: The difference between the free chlorine in supply and in return lines indicates some consumption of free chlorine, as expected, especially in the Main cooling tower, where this difference is higher. Nevertheless the remaining residual is above 0,2ppm free chlorine which is considered to provide sufficient protection
- Soluble iron : the soluble iron is generally low (good). If we look at each tower separately for the data on the date 26/4/22
 - o Main – Sol Fe/Total Fe = $0,10/0,30 = 33\%$
 - o New - Sol Fe/Total Fe = $0,11/0,30 = 37\%$
 - o The values of soluble iron vs total iron are rather high, but acceptable, because the total iron values are low

E. HOT WATER BOILERS – HARDNESS, ALKALINITY & SOLUBLE IRON

The following data were submitted for the period Dec 2021 to April 2022. Averages were calculated by the auditor.

Hot water boiler		
Visit date	Iron (Soluble)	Alkalinity
05-Dec		410
12-Dec		410
21-Dec		410
27-Dec		410
03-Jan		410
09-Jan		410
17-Jan		410
23-Jan		410
31-Jan		410
07-Feb		410
16-Feb		410
20-Feb		410
28-Feb		410
06-Mar		410
14-Mar		410
21-Mar		410
28-Mar	0,10	410
03-Apr		410
10-Apr		410
17-Apr		410
26-Apr	0,1	410
AVERAGES	0,10	410

AUDIT REPORT

Comments

- Soluble iron : the soluble iron is low (good). If we look at the data for the date 26/4/22
 - o Sol Fe/Total Fe = 0,10/0,40 = 25% (acceptable)
- Similar comment as in previous reports: Alkalinity : it is stable, and the calcium hardness is low (5 ppm as CaCO₃ on 26/4/22). According to our previous comments, if there is no leak in the system, the scaling tendency will not affect the system negatively, but if there is a leak then some scaling will be expected (extend of scaling will be proportional to the size of the leak, ie the amount of fresh water introduced in the system).

F. CONSUMPTION OF CHEMICALS

The following data, related to the weekly additions of chemicals to various systems, were submitted for the period Jan 2022 to April 2022. Averages were calculated by the auditor.

Four Seasons Nile Plaza(Consumption)								
Date (weeks)	Corrosshield @ Hot water boiler, liters	Flogard @ Hot water boiler, liters	Continuum @ CT, liters	Spectrus @ CT, liters	Spectrus BD1500, liters	Corrosshield @ Chilled system, liters	Optiguard @ Steam generator, liters	Steamate @ Steam generator, liters
Function/Source of active:	?	?	?	?	Biocide	?	?	?
3 Jan - 9 Jan	0	0	20	10	10	0	4	4
9 Jan - 17 Jan	0	0	55	10	10	0	2	2
17 Jan - 23 Jan	0	0	19	10	10	0	8	8
23 Jan - 31 Jan	0	0	35	10	10	0	3	4
31 Jan - 7 Feb	11	0	32,5	10	10	0	3	4
7 Feb - 16 Feb	0	0	30	10	10	0	3	5
16 Feb - 20 Feb	11	0	47	10	10	0	3	5
20 Feb - 28 Feb	0	0	37	10	10	0	4	6
28 Feb - 6 Mar	0	0	40	10	10	0	3	5
6 Mar - 14 Mar	0	0	40	10	10	0	4	6
14 Mar - 21 Mar	0	0	9	10	10	23	4	6
21 Mar - 28 Mar	0	0	37	10	10	0	5	5
28 Mar - 3 April	0	0	9,25	10	10	0	5	5
3 April - 10 April	0	0	131,25	10	10	0	5	6
10 April - 17 April	0	0	9	10	10	92	4	6
17 April - 26 April	0	0	79	10	10	0	4	5
AVERAGE	1,4	0,0	39,4	10,0	10,0	7,2	4,0	5,1
STANDARD DEVIATION	3,8	0,0	30,6	0,0	0,0	23,3	1,4	1,3

AUDIT R E P O R T

Comments

- Similar comment as in previous reports: As mentioned in previous reports, the actual consumption of chemical (well documented above) should be compared to expected (ie budgeted) consumption per system in order to be able to follow the actual vs budget consumption. This budgeted consumption is missing.
- It will be beneficial to add to the table information about "Function/source of active" for each added chemical. This way the FS staff will easily understand reason for adding the chemical and what is the residual inhibitor (if any) that is related to the addition of the chemical (ie organic phosphate, sulfite, ortho phosphate, nitrite, etc)
- From calculating the averages and standard deviations we can observe that the addition of some chemicals is rather regular (in frequency and quantity) while for other chemicals the addition is irregular (in frequency and quantity). It would be beneficial to add a note explaining the reason for adding the chemical, ie regular weekly addition for biocide, addition due to low residual, etc

G. MICROBIOLOGICAL GROWTH MONITORING ISSUES

The microbiological growth Recording Table was reviewed. Below are the parts of the data that we would like to comment on.

CHILLED WATER SYSTEMS

The following data were submitted for the period Dec 2021 to April 2022.

Comments

- Currently the microbiological growth in Chilled water systems is under control
- Similar comment as in previous reports: For Chilled water systems (closed systems) the control limit on Microorganisms (ie Total Bacteria Count) should be ≤ 1000 cfu/ml (colony forming units/ml). In the previous data we see some measurements exceeding this value and reaching the level of 10.000 cfu/ml. Therefore it is necessary to apply non oxidizing biocide in these systems (like isothiazoline based Spectrus biocide) as soon as the high level of microorganisms is detected.

AUDIT REPORT

System	Date	Microorganisms @22°C/72H	Microorganisms @37°C/48H	Coliforms	E.Coli	Pseudomonas	Legionella
		cfu/ml	cfu/ml	cfu/ml	cfu/ml	cfu/ml	Detection
Control Limits			<100.000	ND in 100MI	ND in 100MI	ND	ND
Chillers – New (P3)	08-01-21	1050	1021	<1 = ND	<1=ND	ND	
Chillers – New (P3)	10-06-21	1.9x10 ⁴	1.5x10 ⁴	<1 = ND	<1=ND		ND
Chillers – New (P3)	01-03-22	20	46	<1=ND	<1=ND		ND
Chillers – New (P3)	04-04-22	65	68	<1=ND	<1=ND		ND
Chiller -Main (P2)	08-01-21	1188	1210	<1 = ND	<1=ND	ND	
Chiller -Main (P2)	10-06-21	1.7x10 ⁴	1.2x10 ⁴	<1 = ND	<1=ND		ND
Chiller -Main (P2)	01-03-22	41	46	<1=ND	<1=ND		ND
Chiller -Main (P2)	04-04-22	72	80	<1=ND	<1=ND		ND
Chillers_Kitchen	08-01-21	1196	1211	<1 = ND	<1=ND	ND	
Chillers_Kitchen	10-06-21	2.1x10 ³	1.5x10 ³	<1 = ND	<1=ND		ND
Chillers_Kitchen	01-03-22	5	8	<1=ND	<1=ND		ND
Chillers_Kitchen	04-04-22	45	60	<1=ND	<1=ND		ND
Chillers_Icemaker	08-01-21	988	1001	<1 = ND	<1=ND	ND	
Chillers_Icemaker	10-06-21	2.3x10 ³	1.7x10 ³	<1 = ND	<1=ND		ND
Chillers_Icemaker	01-03-22	5	9	<1=ND	<1=ND		ND
Chillers_Icemaker	04-04-22	28	25	<1=ND	<1=ND		ND

COOLING TOWERS

The following data were submitted for the period Dec 2021 to April 2022.

Comments

- Currently the microbiological growth in Cooling Tower systems is under control
- Similar comment as in previous reports: For Cooling Tower water systems (open systems) the control limit on Microorganisms (ie Total Bacteria Count) should be ≤ 10.000 cfu/ml (colony forming units/ml). In the data we see some previous measurements approaching this value and reaching the level of >10.000 cfu/ml. Therefore it is necessary to :
 - o Either increase the chlorination
 - o Or apply non oxidizing biocide (like the isothiazoline based Spectrus biocide)
 in these systems as soon as the increasing level of microorganisms is detected.

AUDIT REPORT

System	Date	Microorganisms @22°C/72H	Microorganism ms @37°C/48H	Coliforms	E.Coli	Pseudomonas	Legionella
		cfu/ml	cfu/ml	cfu/ml	cfu/ml	cfu/ml	Detection
Control Limits			<100.000	ND in 100MI	ND in 100MI	ND	ND
Cooling Tower - Main	08-01-21	877	912	<1 = ND	<1=ND		ND
Cooling Tower - Main	22-09-21	750	790	120	<1=ND		
Cooling Tower - Main	10-06-21	240	210	8	<1=ND		ND
Cooling Tower - Main	11-02-21	6.13x10 ³	6.15x10 ³	<1 = ND	<1=ND		
Cooling Tower - Main	12-12-21	240	210	<1 = ND	<1=ND		
Cooling Tower - Main	01-03-22	5	8	<1=ND	<1=ND		ND
Cooling Tower - Main	21-02-22	39	27	<1=ND	<1=ND		
Cooling Tower - Main	06-03-22	80	75	<1=ND	<1=ND		
Cooling Tower - Main	04-04-22	50	28	<1=ND	<1=ND		ND
Cooling Tower - Main	18-04-22	22	30	<1=ND	<1=ND		
Cooling Tower -Kitchen	08-01-21	777	801	<1 = ND	<1=ND		
Cooling Tower -Kitchen	22-09-21	261	294	<1 = ND	<1=ND		
Cooling Tower -Kitchen	10-06-21	2.2x10 ³	1.7x10 ³	13	<1=ND		ND
Cooling Tower -Kitchen	11-02-21	5.16x10 ³	5.2x10 ³	<1=ND	<1=ND		
Cooling Tower -Kitchen	12-12-21	280	300	<1=ND	<1=ND		
Cooling Tower -Kitchen	03-01-22	38	46	<1=ND	<1=ND		ND
Cooling Tower -Kitchen	21-02-22	30	21	<1=ND	<1=ND		
Cooling Tower -Kitchen	06-03-22	10	15	<1=ND	<1=ND		
Cooling Tower -Kitchen	04-04-22	45	80	<1=ND	<1=ND		ND
Cooling Tower -Kitchen	18-04-22	40	42	<1=ND	<1=ND		
Cooling Tower -New	08-01-21	1167	1180	<1 = ND	<1=ND		
Cooling Tower -New	22-09-21	256	289	<1 = ND	<1=ND		
Cooling Tower -New	10-06-21	235	208	8	<1=ND		ND
Cooling Tower -New	11-02-21	8.44x10 ³	8.5x10 ³	15	<1=ND		
Cooling Tower -New	12-12-21	210	202	<1 = ND	<1=ND		
Cooling Tower -New	01-03-22	49	47	<1=ND	<1=ND		ND
Cooling Tower -New	21-02-22	33	27	<1=ND	<1=ND		
Cooling Tower -New	06-03-22	18	12	<1=ND	<1=ND		
Cooling Tower -New	04-04-22	34	40	<1=ND	<1=ND		ND
Cooling Tower -New	18-04-22	35	40	<1=ND	<1=ND		

AUDIT R E P O R T

H. COMMENTS ON THE SERVICE VISIT REPORT (SVR)

The System Technique Service Visit Reports (SVR) for April 2022 were reviewed.

Below please find our comments, related to the last SVR of 26/4/22

Cooling Towers

- No analysis for Kitchen CT
- No analysis for Make-up Water to cooling towers
- No analysis for Alkalinity in all cooling towers
- Control limits of Turbidity were changed in the past to max 10 (report shows 0-0,9)
- Control limits for organic phosphate was changed in the past to 8-10 ppm (no control limit in report)

Sand filters

- No data for turbidity out of filters (therefore it is not possible to calculate efficiency from only the "in" value)
- No data for pressure in/out (therefore it is not possible to calculate differential pressure)

Steam Boiler

- Concentration of iron in boiler water is at the upper limit, there is no discussion of possible causes and recommendations for corrective action(s)
- Conductivity in boiler water is at the lower limit, there is no discussion of possible causes and recommendations for corrective action(s)

Chilled water systems

- Concentration of iron in all systems is at the upper limit, there is no discussion of possible causes and recommendations for corrective action(s)

Swimming Pools

- Control limits of Free Chlorine were changed in the past to 1,0-2,0ppm (report shows 1,0-1,5)

I. PLAN FOR NEXT AUDIT

The next report will be an onsite VISIT. The next visit priorities will be:

- the issues indicated in the last columns of the attached Action Plan ("Auditor 17/4/22" & "Feedback 1/5/22")
- the issues listed in this report (above)

J. ATTACHMENTS & APPENDICES

1. ACTION PLAN- COMMENTS & FEEDBACK